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First-Come,First-Serve CPU Scheduling Algorithm

Write a C program to implement the First-Come, First-Served (FCFS) CPU Scheduling Algorithm. The program should take the number of processes, their Arrival Times (AT), and Burst Times (BT) as input. Calculate and display the Completion Time (CT), Turnaround Time (TAT), and Waiting Time (WT) for each process, along with the Average Turnaround Time and Average Waiting Time.

To evaluate the performance of the algorithm, we use the following formulas:

- **Completion Time (CT):** The time at which a process finishes execution.
- **Turnaround Time (TAT):** Total time taken from arrival to completion.
- $TAT = CT - AT$
- **Waiting Time (WT):** The time a process spends waiting in the ready queue.
- $WT = TAT - BT$

Input Format

1. The first input line contains an integer representing the

Select Language : C (GCC 9.2.0)

```c
    for (int i = 0; i < n; i++) {
        printf("Enter Arrival Time and Burst Time fo
        scanf("%d %d", &at[i], &bt[i]);
    }

    int current_time = 0;
    for (int i = 0; i < n; i++) {
        // If the process hasn't arrived yet, the CP
        if (current_time < at[i]) {
            current_time = at[i];
        }

        // Completion Time = Current Time + Burst Ti
        ct[i] = current_time + bt[i];
        current_time = ct[i];

        // TAT = CT - AT
        tat[i] = ct[i] - at[i];
        // WT = TAT - BT
        wt[i] = tat[i] - bt[i];

        total_tat += tat[i];
        total_wt += wt[i];
    }

    printf("\nAverage Turnaround Time: %.2f", total
```

Test Cases

Run Sample Cases

Run All Cases

Custom Test Case

> Custom Test

Sample Test Case

> Test Case - 1

> Test Case - 2